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APPLIED AI / LLM ENGINEER (RAG, AGENTIC SYSTEMS, LLM ORCHESTRATION)

Applied AI Engineer with 5 years of experience building AI-powered software systems across LLM applications, retrieval pipelines, backend APIs, analytics dashboards, and ML-driven decision tools. Specialized in building agentic LLM systems with multi-step reasoning, tool integration, retrieval pipelines, and production-grade orchestration (RAG + decision loops + async workflows). Strong focus on turning business workflows and unstructured data into usable products using Python, FastAPI, Flask, Streamlit, Docker, and cloud-oriented deployment practices.

Most effective in roles that combine AI product engineering, backend development, model integration, and deployment.

TECHNICAL SKILLS

Languages

Python, JavaScript, TypeScript, SQL, C#

AI / Machine Learning

RAG, Embeddings, Vector Search, FAISS, SentenceTransformers, OpenAI APIs, Prompt Design, XGBoost, Scikit-learn, Model Evaluation

Backend Development

FastAPI, Flask, REST APIs, Async Workflows, .NET

Frontend / Applications

React, Next.js, Streamlit, Dash, Flutter

Data & Processing

PostgreSQL, SQLite, PDF/Text Preprocessing, Feature Engineering, JSON-based Workflows

Infrastructure / DevOps

Docker, Docker Compose, GitHub Actions, AWS, ECS, CI/CD, Logging and Monitoring

Agentic / LLM Systems

Multi-step reasoning, Tool Calling, LLM Orchestration, Prompt Chaining, Decision Loops (familiar with AutoGen / CrewAI)

Keywords

Agentic AI, LLM Systems, Multi-Agent Workflows, Tool Calling, RAG Pipelines, LLM Orchestration, Retrieval-Augmented Generation

PROFESSIONAL EXPERIENCE

AI Engineer

Autosmart Tech

July 2025 – Present

- Full-stack development of mobile and backend applications for automotive workshop

automation, handling customer management, vehicle tracking, job scheduling, and operational analytics.

- Owned end-to-end system design decisions for AI workflows, including trade-offs between latency, cost, and response reliability in production environments.
- Architected and deployed agentic AI workflows (intent parsing → decision routing → backend execution → response generation) using OpenAI APIs serving 15+ active workshops with 200+ monthly service transactions
- Led technical design decisions for voice-first interfaces targeting multilingual (Urdu/Hinglish) mechanics with limited app literacy, achieving 84% intent accuracy through iterative prompt engineering
- Implemented secure API patterns isolating LLM services from direct database access through intent-based JSON boundaries, enabling modular feature addition without expanding AI agent privileges
- Shipped features end-to-end from requirements gathering through production deployment, including real-time job tracking, SLA monitoring, and employee labor-time analytics
- Established code review practices and CI/CD pipelines for mobile releases, reducing deployment friction and ensuring consistent quality across iterative feature rollouts

AI / ML Engineer

Optivex Solutions

Feb 2023 – July 2025

- Built a scalable RAG system for an internal casino knowledge base with role-based access, handling 8TB of continuously growing data across departments.
- Implemented a hybrid offline LLM architecture to ensure client data privacy, sending only vector embeddings to the cloud while keeping sensitive information local.
- Deployed on AWS EC2 with optimized pipelines, leveraging metadata-based caching and precise chunking to reduce latency and minimize computational costs.
- Designed efficient data processing and retrieval workflows, maintaining minimal static tokens while ensuring high query relevance for staff-facing internal systems.
- Engineered the system for scalability and maintainability, enabling real-time natural language querying across multiple roles without compromising security or performance.

ML Engineer/ Data Scientist

Freelance work

Jan 2021 – Feb 2023

- Low product discovery was limiting conversions for an ecommerce client (beauty niche), designed and implemented a hybrid recommendation system (collaborative + content-based) using Python, integrated into the product flow via APIs, which improved user engagement by ~15–20%.
- High customer drop-off was impacting revenue for a subscription-based business, built and deployed a churn prediction model (~80–85% accuracy) using Python (scikit-learn), and enabled targeted retention efforts based on at-risk user segments.
- Manual data analysis was slowing decision-making, built end-to-end data pipelines and ML workflows (Python, FastAPI) covering preprocessing, training, and API-based inference, reducing analysis turnaround time by ~25–30%.
- Inconsistent model performance across use cases was affecting reliability, improved model stability through feature engineering, hyperparameter tuning, and validation strategies, increasing key metrics by ~10–15%

KEY PROJECTS

Autonomous Agentic Research Assistant

Tech: Python, FastAPI, FAISS, OpenAI APIs, AsyncIO

- Built an agentic LLM system with tool-calling, retrieval orchestration, and confidence-based decision loops, enabling evidence-grounded conversational responses with dynamic fallback and refusal mechanisms.
 - Designed multi-step decision pipeline (plan → retrieve → evaluate → respond) with adaptive execution based on confidence, async orchestration, and failure-safe retries.
 - Improved reliability in a 50-query regression harness: runtime errors 12% → 0% (6/50 → 0/50) and clean retrieval 88% → 100% (44/50 → 50/50) while maintaining ~2.3s/query average throughput.
 - Reduced tail latency with async parallel retrieval + cancellation under a strict 7s budget; achieved warm p95 /chat latency 5.21s (median 3.70s) on the measured multi-turn suite, and removed first-user cold-start impact via prewarm + [/ready](#) readiness gating.
 - Added anti-hallucination controls: relevance filtering/scoring + confidence gating so the system visibly degrades (“insufficient relevant sources”) rather than generating confident, unrelated answers.
 - Hardened external API usage for production by integrating Semantic Scholar auth and enforcing 1 req/sec rate limiting, reducing 429/rate-limit failures under real traffic patterns.
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Identity Theft Protection (IDP MasterCard)

Tech: .NET, Angular, REST APIs, SFTP, Webhooks, Azure/AWS-style distributed systems, Background Jobs, Event-Driven Architecture

- Built an end-to-end DCB subscription orchestration platform (.NET + Angular), integrating telcos via APIs, webhooks, vouchers, and SFTP; automated provisioning into Mastercard systems via IRIS APIs.
 - Engineered idempotent event ingestion and correlation (MO/MT/DR) to handle duplicate events, retries, and out-of-order delivery, ensuring correctness under real-world failure scenarios.
 - Designed fault-tolerant lifecycle workflows (activation, renewal, deactivation) with background jobs and partner-specific retry strategies, maintaining consistency across billing and provisioning systems.
 - Implemented secure PII handling (MSISDN encryption at rest) and hardened partner integrations with signed requests and authentication layers; established end-to-end observability using correlation IDs.
 - Developed an AI-driven reconciliation anomaly detection module to identify mismatches between internal events and partner settlement reports, cluster root causes, and prioritize investigations via ranked alerts.
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Casino Knowledge Base RAG System

Tech: Python, FAISS, BM25, SentenceTransformers, Cross-Encoders, OpenAI APIs, AWS EC2, AsyncIO, Vector Databases, Caching Systems

- Built a scalable RAG system for an 8TB internal knowledge base across HR, finance, and operations, with role-based access control for staff.
- Implemented a hybrid search pipeline combining BM25 + vector embeddings for efficient and relevant document retrieval.
- Applied query rewriting and expansion to improve retrieval coverage and relevance, followed by cross-encoder re-ranking to ensure high-precision results.
- Engineered a hybrid offline LLM architecture to maintain privacy: only vector embeddings are sent to the cloud while raw data remains local.
- Deployed on AWS EC2, optimized with metadata-based caching, precise chunking, and minimal static tokens to reduce latency and compute costs.
- Designed continuous data ingestion and retrieval workflows for incremental updates while preserving accuracy and scalability.
- Ensured secure, modular access, separating LLM query handling from database privileges, enabling safe expansion of intents and queries.
- Extended RAG pipeline into semi-agentic retrieval system with query rewriting, dynamic routing, and relevance-driven execution paths.

AutoInsight (Digitilizing Workshops)

Tech: Flutter, .NET, OpenAI APIs, FastAPI-style backend patterns, PostgreSQL, REST APIs, TTS/STT pipelines, Role-based Access Control

- Built AutoInsight, a Flutter + .NET workshop operations app that digitizes repair workflows, replacing paper-based tracking for customers, vehicles, jobs, and employee assignments.
- Designed relational workflow logic where one customer manages multiple vehicles, each vehicle supports multiple timed service jobs (oil change, tuning) with live status tracking and job-specific SLAs.
- Implemented employee job execution with pause/resume timers for accurate labor-time tracking and real-time operational visibility across the workshop floor.
- Architected a voice assistant using OpenAI (intent extraction → backend query → response generation → TTS) enabling natural-language status queries in Urdu/Hinglish without exposing database access to LLM services.
- Enforced security boundaries between AI and data layers: LLM services interact only through intent-based JSON APIs while backend owns database access, enabling modular intent addition without expanding LLM privileges.
- Achieved 84% voice-query accuracy through iterative prompt engineering and manual validation across 50+ garage-management intent scenarios.

EDUCATION

ADP in Artificial Intelligence

Superior University

CERTIFICATIONS

- Large Language Models on AWS
- MLOps Essentials: Model Deployment and Monitoring
- MLOps Essentials: Model Development and Integration
- MLflow and Hugging Face Tools
- AWS Machine Learning / Deployment Coursework